
A Statistical Audit of LLM Calibration Heterogeneity Across Knowledge Domains

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Abstract

1 LLM calibration is nearly always reported as an aggregate scalar, obscuring where
2 miscalibration concentrates. We develop a statistical audit framework (three-level
3 mixed-effects model, isotonic reliability curves, Benjamini–Hochberg FDR) and
4 apply it to MMLU across seven instruction-tuned models from three families
5 spanning six parameter scales. Subject-level ECE rankings are highly correlated
6 (Spearman $\rho \in [0.80, 0.94]$ across all 15 pairs of larger models), the same subjects
7 are robustly worst- and best-calibrated across models, and within Qwen the entropy
8 coefficient on the calibration gap grows monotonically with scale. The stability of
9 these patterns provides empirical motivation for subject-aware recalibration.

10 1 Setup

11 We extract token-level logprobs over A/B/C/D for all 14,042 MMLU questions on GPT-4o-mini,
12 Llama-3-8B-Instruct, and Qwen-2.5- $\{0.5B, 1.5B, 3B, 7B, 14B\}$ -Instruct. The outcome is the signed
13 calibration gap $\text{gap}_i = c_i - y_i$ (max softmax confidence minus correctness). A three-level mixed
14 model $\text{gap}_{ijk} = \mu + \mathbf{x}_{ijk}^\top \boldsymbol{\beta} + u_k + v_{jk} + \varepsilon_{ijk}$ decomposes variance across questions, subjects, and
15 domains; per-subject gaps are tested with one-sample t -tests and Benjamini–Hochberg FDR.

16 2 Findings

17 **(i) Where calibration fails—and where it doesn’t.** Across the six larger models, virology and
18 professional_law appear in the top-10 most miscalibrated for every model; a cluster of college-
19 level technical subjects (abstract_algebra, college_mathematics, machine_learning) is in
20 the top-5 for four of six. Symmetrically, marketing and high_school_psychology are in the
21 bottom-10 for every model. Specialization level tracks miscalibration: high-school subjects show
22 systematically smaller gaps than their college-level counterparts.

23 **(ii) Cross-model and cross-family stability.** All 15 pairwise Spearman correlations among the six
24 larger models fall in $[0.80, 0.94]$ (Figure 1); cross-family GPT-4o-mini vs. Qwen-7B/14B at 0.92/0.91
25 is comparable to the strongest within-family pair (Qwen-7B vs. Qwen-14B at 0.94). Qwen-0.5B is a
26 conspicuous exception ($\rho \in [0.13, 0.20]$), driven by $10.6\times$ positional bias toward the D option—we
27 read this as a lower bound below which our elicitation protocol breaks down.

28 **(iii) Within-Qwen scaling.** Across four matched-precision Qwen checkpoints (1.5B fp16, 3B, 7B,
29 14B), the entropy coefficient on the calibration gap grows monotonically ($-0.004 \rightarrow +0.006 \rightarrow$
30 $+0.029 \rightarrow +0.040$, last two $p < 0.001$): at 7B and 14B, confidence stays near saturation across
31 the first three entropy quartiles while accuracy still drops, so the gap widens on the model’s harder
32 items (Q1 vs. Q4 gap: $+0.05$ vs. $+0.37$ at 7B; $+0.03$ vs. $+0.46$ at 14B). The between-subject ICC

33 jumps from 1.5B to 3B ($0.017 \rightarrow 0.051$) and is essentially flat thereafter. This is consistent with prior
 34 reports of RLHF-induced confidence saturation.

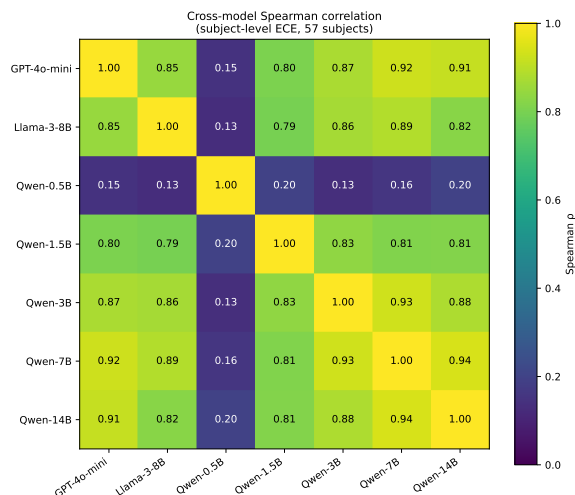


Figure 1: Spearman rank correlation of subject-level ECE across the seven models (57 common subjects). Among the six larger models, all 15 pairwise correlations fall in $[0.80, 0.94]$. Qwen-0.5B’s correlations with the others are $[0.13, 0.20]$, reflecting a qualitatively different miscalibration pattern driven by extreme positional bias.

35 3 Implications

36 Cross-model stability is the precondition that makes subject-aware recalibration worth scoping: a
 37 correction informed by one model’s subject rankings will generalize to others. A global recalibration
 38 would under-correct the worst-calibrated cluster (professional_law, virology, college-level
 39 technical content) while over-correcting subjects like marketing. The robustly worst-calibrated
 40 subjects double as a transferable diagnostic for flagging miscalibration on new models before
 41 deployment, without re-running the full 57-subject audit.